

Unveiling measurement precision of Indonesian version DASS-21: A rating scale model

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Abstract

Testing on the structure factor of the Indonesian version of Depression Anxiety Stress Scale (DASS)-21 has been carried out using confirmatory factor analysis. To prove the quality of the psychometric properties of DASS-21 in more detail, this study aims to evaluate the internal validity and reliability of the Indonesian version of DASS-21 using the Rasch rating scale model. This research approach uses a cross-sectional design, and to get as many as 424 undergraduate students age ranged from 17 to 26 years old (Mage = 20.49 and SD = 1.79) in various regions in Indonesia, the researcher uses convenience sampling. The results show that the Indonesian version of DASS-21 is unidimensional even though it is tested on each subscale or as a whole. It has an excellent reliability index, all items used are fit, and the use of four answer choices is appropriate, but identified as many as 6 of the 21 items detected experiencing a differential item functioning that shows the weaknesses of the Indonesian version of DASS-21. Psychometrically, only 15 of the 21 items from the Indonesian version of DASS-21 are recommended for measuring psychological distress in Indonesia.

Keywords

Differential item functioning (DIF), Indonesian version DASS-21, psychometric properties, rasch rating scale model

Introduction

Psychological health problems frequently occur across various segments of society in many countries. One common form of psychological health issue experienced by individuals is stress in carrying out daily activities (Monteiro *et al.*, 2014). The situation became more pronounced during the pandemic a few years ago, when psychological health issues received greater attention due to the emergence of numerous psychosocial and mental health problems (Mukhtar, 2020). Research has also revealed that mental health problems are widely experienced by university students (Di Consiglio *et al.*, 2021), where three main sources of stress among students have been identified: excessive academic workload, pressure to succeed after graduation, and the preparedness to pursue life plans upon completing their studies (Beiter *et al.*, 2015).

In addition to stress, university students often experience anxiety and depression. Previous studies have reported that depression, anxiety, and stress are among the most prevalent forms of psychological disorders (Shea *et al.*, 2009). In psychology, these three conditions are collectively referred to as psychological distress. The impacts of psychological distress on students may include difficulties in sleeping and physical health problems (Liu *et al.*, 2020). Moreover, students may experience a sense of inability to complete their academic tasks (Fradelos *et al.*, 2019). Consistently, Stanton *et al.* (2020) found that individuals experiencing psychological distress are more likely to engage in negative behavioral changes, such as decreased immune functioning and engaging in unusual behaviors like consuming alcohol and smoking (Stanton *et al.*, 2020).

A phenomenon that commonly occurs due to psychological distress among university students is suicidal ideation

(Bantjes *et al.*, 2016). Students who are unable to cope with psychological distress are highly likely to face difficulties in dealing with the problems they encounter (Houston, 1971). Moreover, students who are just beginning their studies at the university level inevitably face numerous challenges during the transition to an independent environment with many responsibilities to manage (Kaloeti *et al.*, 2019). Therefore, the prevalence of psychological distress has become an important issue to investigate, as its rate continues to increase among students (Rith-Najarian *et al.*, 2019).

A study conducted by Teh *et al.* (2015) on 397 undergraduate students at Melaka Manipal Medical College in Malaysia found that the students experienced psychological distress at moderate to extreme levels, with prevalence rates of depression (30.7%), anxiety (55.5%), and stress (16.65%). Similar findings were reported among 1,074 students at San Jorge University (SJU) from the Faculty of Health Sciences, Faculty of Communication Sciences, and Faculty of Architecture, which showed moderate prevalence rates of depression (18.4%), anxiety (23.6%), and stress (34.5%) (Ramón-Arbués *et al.*, 2020). In the general population, anxiety had the highest prevalence, with 41.3% of 51,547

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individuals requiring treatment for psychological distress (Alonso *et al.*, 2018).

Psychological distress remains a major public health concern, especially among young people and university students. Recent national report released after the pandemic have highlighted a noticeable increase in mental health complaints in Indonesia. For example, Kemenkes (2023) report rising consultation related to depression, anxiety, and stress among adolescent and young adults. Likewise, UNICEF (2024) post-pandemic assessment show that Indonesian youth continue to experience heightened level of emotional distress, academic pressure, and adjustment difficulties, all of which contribute to greater vulnerability to psychological problem.

The importance of addressing psychological distress should not be underestimated, as the increasing prevalence of psychological distress has serious consequences for mental health (Dalky & Gharaibeh, 2019). Appropriate interventions can provide significant benefits to the lives of those affected. Several studies have identified suitable treatments to manage psychological distress. Habibi *et al.* (2013) for instance, proposed a “stress management skill” program to address stress-related problems. Other researchers have suggested mindfulness as an effective approach to cope with psychological distress (Flett *et al.*, 2020). Even financial resources can serve as a solution to psychological distress among individuals with lower socioeconomic status (Dakin & Wampler, 2008). Additional techniques scientifically introduced to address psychological distress in students include mood management skills and coping self-efficacy (Stallman *et al.*, 2018).

In addition to treatment, an important aspect in understanding psychological distress among students is the process of initial screening. Routine screening should be carried out through assessments that measure the levels of depression, anxiety, and stress among students (La Torre *et al.*, 2022). Such assessments serve as an initial screening step that can later be used to provide early interventions. A study conducted by Kartika *et al.* (2020) in Indonesia also emphasized the importance of mental health first aid within educational settings, both in schools and universities. This highlights the necessity of using appropriate psychological instruments to measure psychological distress for early screening.

Several previous studies have identified various instruments for measuring depression, anxiety, and stress. For instance, well-known instruments for measuring depression include the Beck Depression Inventory (Beck *et al.*, 1988) and the Center for Epidemiologic Studies Depression Scale (González *et al.*, 2017). Commonly used instruments for measuring anxiety include the Hamilton Anxiety Scale (Maier *et al.*, 1988) and the Clinical Anxiety Scale (Snaith *et al.*, 1982). The Perceived Stress Scale has been widely applied in research as an instrument specifically designed to measure stress (Siqueira Reis *et al.*, 2010). However, each of these instruments focuses on only one domain of measurement. The Depression Anxiety Stress Scale (DASS) (Lovibond & Lovibond, 1995) is the only instrument capable of measuring these forms of psychological distress as a single construct and simultaneously.

From a psychometric perspective, the DASS offers several advantages. It is an instrument with high stability and responsiveness compared to other measures (Yohannes *et al.*, 2019). The reliability values obtained from the DASS

are consistently above .80, indicating that the instrument is reliable (Gallagher *et al.*, 2014). Another strength lies in its factor structure, which generally represents psychological distress and is suitable for Western cultural contexts (Mellor *et al.*, 2015). However, in addition to these strengths, the DASS also has limitations, particularly in the items designed to measure stress, which are considered insufficient to capture stress in a broader sense (Jovanović *et al.*, 2021).

The DASS has previously been tested in Indonesia, with construct validity examined using Confirmatory Factor Analysis (CFA) in 2020 (Widyana, 2020) and further analysis conducted with the Rasch Model in 2022 (Ifdil *et al.*, 2022). Although the DASS-21 has demonstrated cross-cultural applicability in various countries, psychological instrument do not automatically maintain equivalent function across different linguistic and cultural contexts. Cultural adaptation remains necessary because emotional expression, idioms of distress, and symptom interpretation can differ substantially between populations, potentially altering how individual respond to specific items. Therefore, a rigorous psychometric evaluation-particularly using Rasch model is required to assess item functioning, rating scale performance, and measurement fairness.

Unlike CFA, which focused primarily on factor structure, the Rasch approach provided detailed item-level diagnostics identifies differential item functioning across demographic groups and evaluates whether response categories operated as intended. However, earlier Indonesian studies did not provide sufficient insight into whether the DASS items function fairly equivalently, whether the response ratings used in the DASS are appropriate, or whether the items exhibit different levels of difficulty across subscales. Due to the lack of comprehensive information regarding the psychometric quality of the DASS, the present study was conducted with the aim of evaluating the instrument more thoroughly from a psychometric perspective.

Rasch Rating Scale Model

The Rasch model was introduced by the mathematician George Rasch in 1960. It was initially developed to analyze dichotomous data (Kreiner, 2013). The Rasch model is a statistical modeling approach that demonstrates the potential relationship between person parameters and item parameters. Bond & Fox (2015) explain that person parameters refer to an individual's level of ability, while item parameters indicate the level of item difficulty. In principle, the Rasch model is primarily used to measure the degree of item difficulty in relation to the ability level of individuals.

In its development, the Rasch model has also been applied to polytomous data. In particular, polytomous measurement (in this case, response categories) often lacks equal intervals because it is ordinal in nature. Therefore, the Rating Scale Model (RSM) or the Partial Credit Model (PCM) was introduced as appropriate techniques for measuring attitudes (Andrich & Marais, 2019). Analysis using the Rasch model requires a fundamental assumption to be met, namely unidimensionality (Mair, 2018). In addition, several criteria must be considered, such as infit Mean-Square (MNSQ) and outfit Mean-Square (MNSQ), with outfit MNSQ being the most frequently applied criterion in Rasch studies.

Analysis using the Rasch model has the advantage of providing more detailed information, including fit statistics,

reliability values, separation indices, comparisons between individual ability levels and item difficulty, as well as the detection of potential bias across groups, which in the Rasch model is referred to as differential item functioning. In addition to producing credible analytical results (Mohd Matore *et al.*, 2021), Jong *et al.* (2015) also argue that the Rasch model is a highly accurate method for precisely determining the quality of an instrument.

Method

Study Design, Participants and Procedure

This study employed a cross-sectional design, in which data collection was conducted within a single time frame (Olsen & George, 2004). The cross-sectional design is also an appropriate research design for assessing both physical and psychological health. The participants consisted of 424 undergraduate students from various regions in Indonesia, recruited through convenience sampling. The undergraduate students recruited for this study came from various universities across Indonesia, including both public and private higher education institutions, and represented multiple fields of study such as psychology, education, engineering, health sciences, and social sciences. The inclusion criteria required participants to: (1) be currently enrolled as undergraduate students, (b) between 17 to 26 years old, and (3) provide informed consent to participate. The criteria ensured that sample reflected academically representative group of Indonesian undergraduate students.

The sample size determination followed the recommendation of Walpole *et al.* (2016), which states that when the exact population size of a sample is unknown, the minimum required sample for research is > 385 participants. Data were collected using two methods: web-based and paper-and-pencil. All participants were identified as university students, informed of the purpose of the study beforehand, provided informed consent, and voluntarily agreed to participate. This study obtained ethical clearance from the Indonesian Scientific Psychology Consortium (Konsorsium Psikologi Ilmiah Nusantara).

Measure

The instrument used in this study was the Depression Anxiety Stress Scale (DASS), developed to measure psychological distress in individuals (Lovibond & Lovibond, 1995), in which each individual is asked to report their condition over the past week. This study employed the short version, DASS-21, obtained from the official website (<https://www2.psy.unsw.edu.au/dass/>). The DASS-21 consists of three subscales, each comprising seven items measuring depression, anxiety, and stress. Each item is rated on a four-point response scale (0 = Never, 1 = Sometimes, 2 = Often, 3 = Usually). In general, the DASS-21 items focus on mood, self-esteem, outlook on the future, fear, physiological arousal, and feelings of tension. The total score of the DASS-21 is calculated by summing the item scores to represent an individual's level of psychological distress. The results of validity and reliability analyses are presented in the Results section.

The Indonesian version of DASS-21 used in this study has previously undergone an establish cultural adaptation

procedure, including forward translation, back-translation, expert judgement, and evaluation of sematic and conceptual equivalence, as documented in earlier validation study (e.g. Ifdil *et al.*, 2022; Widyana, 2020)).

Statistical analysis

This study used Winstep version 3.65 for analysis. Analytic logic the Rasch model allowed for the evaluation of item fit, the functioning of rating scale quality, reliability and separation indexed for both persons and items, as well as fairness across gender groups through differential item functioning (DIF). Rasch Rating Scale Model (RSM) was selected because all items in the DASS-21 share an identical four-point response format, RSM is specifically designed for instruments in which every items uses the same set of response categories.

The statistical criteria applied included testing the assumption of unidimensionality by examining the Variance Explained by Measure, with a threshold of $> 40\%$ (Holster & Lake, 2016). The criteria for item separation and person separation indices were set at > 3 (Duncan *et al.*, 2003). To evaluate the appropriateness of the response categories, the logit difference between response options was required to fall within the range of 1.4–5 logits (Zile-Tamsen, 2017). Internal consistency was assessed using Cronbach's alpha coefficient, with values above .80 indicating good reliability (Gallagher *et al.*, 2014). For item fit, the acceptable range of Outfit Mean-Square (MNSQ) values was .50–1.5 (Boroel *et al.*, 2017). The focus of the Rasch model analysis was to determine item difficulty, with acceptable values ranging from $-.30$ to $+.30$ (Wicaksono *et al.*, 2021). Finally, differential item functioning (DIF) analysis was conducted, with a DIF contrast between .40–.60 indicating a moderate level and between .60–.80 indicating a high level (Rogers & Swaminathan, 1990).

Result

Descriptive Statistic

Data integrity checks were performed prior to conducting the Rasch analysis. There procedure included examining the dataset for missing value, identifying univariate and multivariable outliers, and evaluating response patterns such as straight-lining of inconsistent category use. The inspection that no missing value were present, all participants fell within acceptable in threshold, and there were no indication of careless, random, or inattentive responding. Therefore, the full dataset consisting of 424 participant was deemed appropriate and retained for subsequent Rasch analysis. A total of 424 undergraduate students from various regions in Indonesia participated in this study. The participants' ages ranged from 17 to 26 years ($M = 20.49$, $SD = 1.79$). Female participants accounted for 59.2%, which was higher than the proportion of male participants.

Model Fit Statistic DASS-21

Table 1 presents the overall model fit statistics for the DASS-21. The analysis was conducted for each subscale as well as for the entire set of items, allowing for comparison. The results showed that both the subscales (depression, anxiety, and stress) and the full scale (psychological distress) met

the assumption of unidimensionality. Based on the model fit indices, significant Chi-square values were observed.

Another finding that can be interpreted is the comparison between the item measure mean and the person measure mean. The person measure mean in the stress subscale was higher than the item measure mean, indicating that the participants experienced higher levels of stress. In contrast, the depression and anxiety subscales showed lower levels in terms of quantity. When viewed as a whole, the person measure mean for psychological distress was lower than the item measure mean, suggesting that the participants overall did not experience excessive psychological distress.

The study also identified differences in anxiety levels based on gender. With a *t*-value of -2.253 and a significance level of $2.5\% < 5\%$, the results indicated that female participants experienced higher anxiety than male participants. The reliability values also met the recommended thresholds, both for each subscale and for the overall scale. This suggests that the DASS-21 can be applied either as a unidimensional model for the full scale or as a multidimensional model for each subscale.

Furthermore, for the person–item separation index, only the model using all items (psychological distress) met the threshold of > 3 , indicating that the participants were representative of the broader population. When analyzed by subscale, the participants appeared homogeneous. This suggests that the sampling technique used was appropriate for measuring psychological distress among students. The item separation index exceeded the threshold, demonstrating that the items were representative in capturing psychological distress. In conclusion, based on the Rasch analysis, the DASS-21 is an appropriate instrument for measuring psychological distress, as all statistical criteria were satisfied

Item Fit and Wright Map

Wright map provides a visual representation the distribution of the item difficulty and participant ability on the same logit scale. On the left side of the map, each “#” symbol represent approximately three participants. The distribution shows that most participant fall between -1 to $+1$ logit, indicating that the majority of students report low to moderate levels of psychological distress. Very few participants are located at the extreme upper and lower ends of the scale, suggesting that severe or very minimal symptoms are relatively uncommon in this sample. On the right side, the DASS-21 items are arranged according to their estimated difficult levels. Items located toward the logit lower values (e.g., A4, S7, S4, S1, S3) are easier to endorse, meaning these symptoms are more commonly experienced among participants. Items placed at higher logit positions (e.g., A2, D7) are more difficult to endorse, indicating that there symptoms reflect more severe or less frequently experienced distress.

Overall, the Wright Map in figure 1 demonstrates the the range of items difficulties corresponds reasonably well with the range of participant distress level. However distribution suggest that, for this non clinical student population, most DASS-21 item reflect symptoms at the mild to moderated level, with only a few item capturing higher levels of distress. This indicated that instrument adequately represent the psychological distress characteristics of the sample while maintaining meaningful variability across items.

Table 2 present the calibration of each item in the DASS-21. All items can be considered fit, as they fell within the specified range of Outfit MNSQ values. Based on categorization thresholds, the DASS-21 items can be grouped into three categories: difficult (items D7, D1, D6, D5, A2, A6, S6), medium (items A3, A1, A5, A7, S5, S2), and easy (items D3, D4, D2, A4, S7, S4, S1, S3). The Wright Map in Figure 1 provides a clear visualization of the distribution of item difficulty levels.

Rating Scale Diagnostics

One of the advantages of using the Rasch model is the ability to determine whether the response categories of an instrument are appropriately structured. The DASS-21 employs four response options (never, sometimes, often, and usually). The analysis showed that these four alternatives met the threshold requirements. For instance, the threshold difference between “sometimes” and “often” was 1.89 logits, indicating that the response options were properly ordered and functioned as intended. The response patterns of participants, as illustrated in Figure 2, confirmed that all response categories were represented and formed an appropriate distribution.

DIF Analysis

The Rasch model analysis also enabled the detection of item bias within the instrument, commonly referred to as Differential Item Functioning (DIF). The DIF analysis identified six items in the DASS-21 that exhibited DIF. Two items (D5 and S7) favored male participants, while four items (A3, A4, A7, and S3) favored female participants. All six items detected with DIF were classified at moderate to high levels.

Discussion

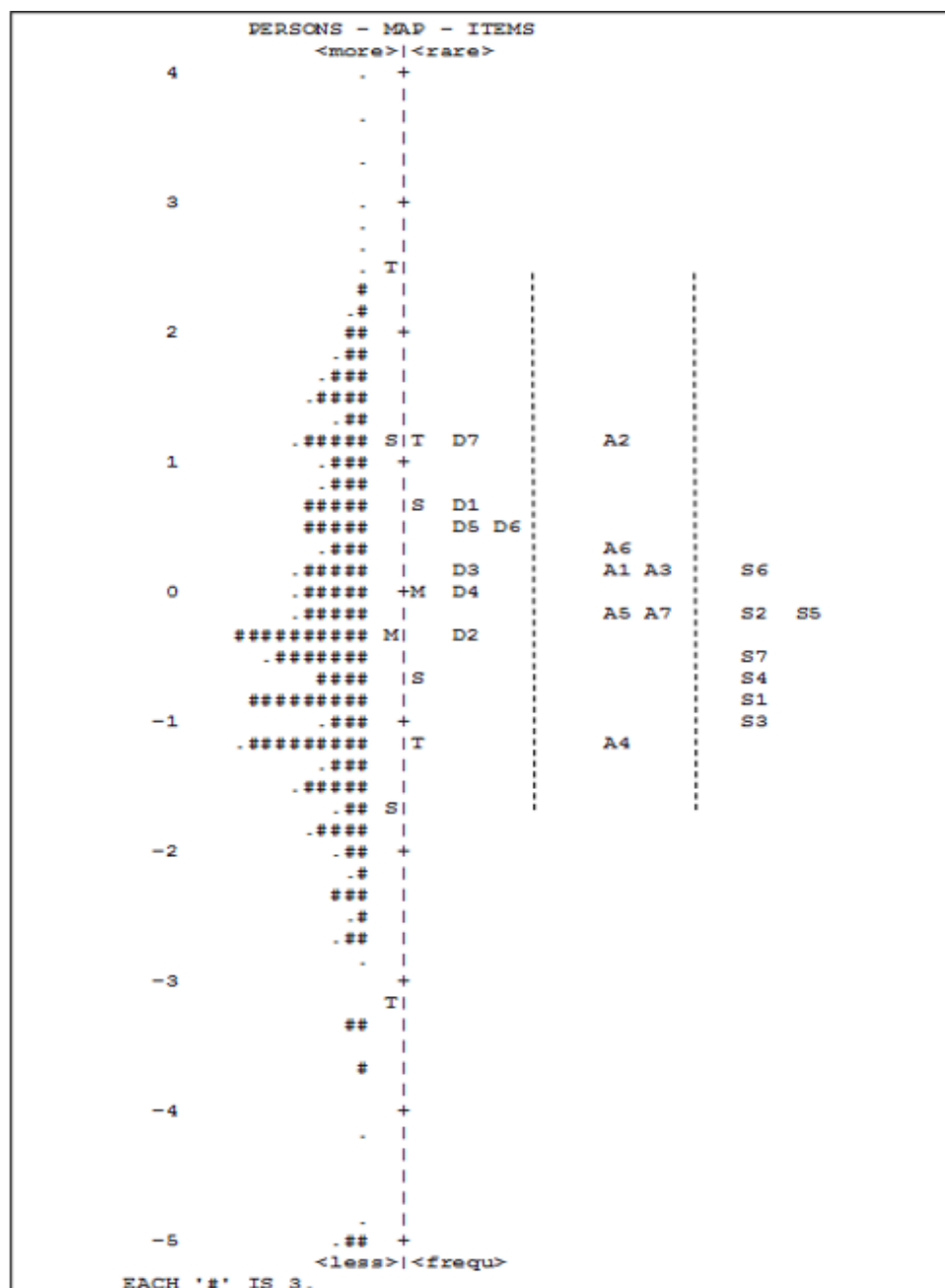
This study revealed several important findings that warrant discussion. The unidimensional model that met the statistical criteria for testing with the DASS-21 was psychological distress, indicating that the DASS-21 can measure depression, anxiety, and stress as a unified construct. This finding contrasts with previous research that analyzed the DASS using a multi-factor model (Widyana, 2020). The results of this study suggest that, in principle, the DASS can be applied both as a whole instrument (based on Rasch model analysis) and as subscales (based on CFA).

The descriptive statistics showed that most students reported distress level within the mild to moderate range, the person measure mean value (person ability (θ)) was lower than the item measure mean (average item difficulty (b)), indicating that participants show lower distress that required to endorse the mode difficult symptoms. This pattern is consistent with non-clinical sample, but this finding contradicts previous research suggesting that mental health problems are widely experienced by university students (Di Consiglio *et al.*, 2021). However, a closer examination of the subscales revealed that female students in Indonesia reported higher levels of anxiety compared to male students. This result is consistent with previous studies indicating that women are more vulnerable in certain situations and therefore more likely to experience higher levels of anxiety than men (Khoshaim *et al.*, 2020; Wang & Zhao, 2020).

Table 1. Model Fit Statistics, Reliability and T-Test.

Action	Model Fit	Raw Variance	Item Measure M (SD)	Person Measure M (SD)	Person Separation Index	Item Separation Index	Reliability	Significant t-test
DASS-21-Depression 7-items	$\chi^2 = 4841.17$ $p < .01$	60.2%	.00 (.59)	-.76 (1.89)	2.36	6.78	.90	94.2%
DASS-21-Anxiety 7-items	$\chi^2 = 5309.57$ $p < .01$	57.9%	.00 (.70)	-.31 (1.58)	2.16	8.95	.87	2.5%
DASS-21-Stress 7-items	$\chi^2 = 5302.22$ $p < .01$	53.9%	.00 (.43)	.14 (1.62)	2.16	5.20	.87	50.1%
DASS-21-Psychological Distress 21-items	$\chi^2 = 16919.88$ $p < .01$	53.1%	.00 (.61)	-.29 (1.43)	3.80	7.96	.95	31.6%

Note. SD = Standardized Deviation, χ^2 = Chi-square

**Figure 1.** Wright Map Indonesian Version of DASS-21.

The Rasch analysis for the psychological distress model met the threshold for the person separation index, indicating that the sampling technique used in this study was appropriate

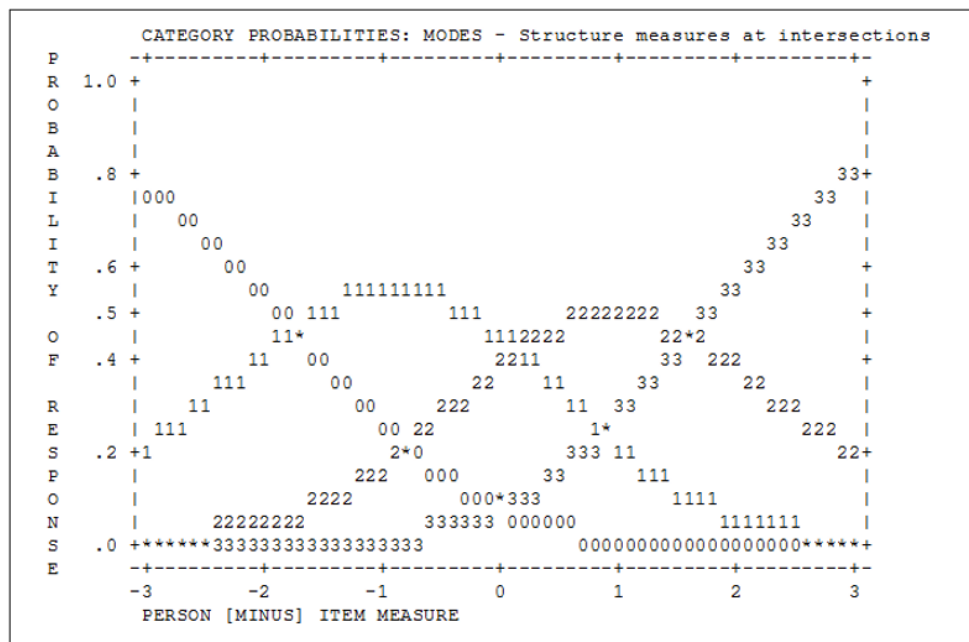
and accurately represented the student population in Indonesia. Although convenience sampling is widely accepted in early stage psychometric studies, the generalization of the present

Table 2. Item Fit Statistics for Indonesian Version DASS-21 Subscales.

Item Number	Location Value	Standart Error	Outfit MNSQ	PTMEA Correlation
DASS-21-Depression				
D7	1.24	.08	.92	.66
D1	.63	.07	.81	.71
D6	.53	.07	.95	.70
D5	.45	.07	1.14	.63
D3	.11	.07	1.07	.68
D4	.05	.07	.70	.77
D2	-.30	.07	1.14	.66
DASS-21-Anxiety				
A2	1.16	.08	1.19	.61
A6	.41	.07	1.09	.66
A3	.16	.07	1.14	.68
A1	.09	.07	1.36	.57
A5	-.16	.07	.68	.77
A7	-.19	.07	.89	.75
A4	-1.17	.07	1.03	.72
DASS-21-Stress				
S6	.11	.07	.98	.66
S5	-.12	.07	.77	.74
S2	-.18	.07	1.12	.63
S7	-.44	.07	1.30	.60
S4	-.67	.07	.65	.79
S1	-.76	.07	1.06	.68
S3	-.96	.07	.98	.73

Table 3. Rating Scale Categories Analysis.

Categorization Response	Observed Count	Percentage (%)	Observed Average	Rating Scale Threshold	Standart Error
0 – Never	1892	22%	-1.87	None	None
1 – Sometimes	3129	36%	-.66	-1.74	.03
2 – Often	2389	27%	.40	.15	.03
3 – Usually	1305	15%	1.59	1.59	.04

**Figure 2.** Category Probabilities Indonesian version of DASS-21.

findings should be approached cautiously. Future research is recommended to employed probability sampling or draw

participants from a broader range of universities with more balance demographic characteristics.

Table 4. DIF by Gender.

Item Number	DIF		DIF Contrast	t	Prob
	Male	Female			
D5	.61	.19	.42	2.83	.0056
A3	-.01	.44	-.45	-2.98	.0119
A4	-1.43	-.77	-.66	-4.44	.0004
A7	-.44	.22	-.65	-4.40	.0000
S3	-1.13	-.70	-.43	-2.88	.0490
S7	-.22	-.78	.56	3.83	.0006

Note. SD is Standardized Deviation, χ^2 is Chi-square

The highest reliability value was also obtained for the psychological distress model, with a coefficient of .95. This finding supports previous research showing that the reliability values of the DASS are consistently above .80, confirming that the instrument is reliable (Gallagher *et al.*, 2014). The Rasch item fit analysis analysis demonstrated that all items fell within acceptable infit and outfit MNSQ ranges. This indicated that the items work together to measure single underlying construct of psychological distress. Good model fit also suggests that the DASS-21 structured is functioning coherently and that no item behaves unpredictably in relation to the latent trait.

The rating scale diagnostic further that the four response categories functioned appropriately. The thresholds were ordered, and each category showed adequate differentiation, demonstrating that participants were able to consistently distinguish between increasing levels of distress severity. This finding supports the appropriateness of the DASS-21 response format and indicates that no collapsing or restructuring of categories is required.

A closer examination revealed that the depression subscale contained more items with higher difficulty levels (items that were less likely to be endorsed by participants) compared to the other subscales. Four items were identified as having high difficulty levels, which may indicate that the participants did not experience depression at the time of data collection. As a result, many participants were less inclined to endorse the items in the depression subscale. For the items in the stress subscale, none were found to have high difficulty levels. All items in this subscale were more frequently endorsed by participants. This finding is consistent with previous research indicating that one limitation of the DASS is that the stress subscale items do not adequately capture stress in a general sense (Jovanović *et al.*, 2021), as the items in this subscale were too easily endorsed by the participants.

Another important finding that warrants discussion is the detection of six items with DIF in the Indonesian version of the DASS-21. Previous studies using the Indonesian version of the DASS-21 have not reported whether gender bias was present. The items D5, A3, A4, A7, S3, and S7 favored one group over the other, either female or male participants. This finding contrasts with psychometric research that emphasizes the importance of measurement equivalence in psychological assessment (Raju *et al.*, 2002). When an item favors one group, it can be assumed to lack equivalence and may be less suitable for measurement procedures.

Six items (D5, A3, A4, A7, S3, and S7) were flagged for DIF by gender. Examination of item wording shows that several items describe emotional states that may be interpreted

differently across gender groups. For instance, items such as “*saya merasa tidak berguna sebagai manusia*” (D5) and “*saya merasa sering kehabisan tenaga akibat perasaan cemas*” (A3) may resonate differently depending on cultural norms surrounding emotional expression. In Indonesian culture, female students may more socially permit to express fear or sadness, whereas male students may interpret such items in a more restrained manner, contributing to DIF.

Similar stress items like S3 and S7 include cognitive components (e.g., difficulty relaxing) that may be experienced or described differently across genders due to cultural expectations about coping. Cultural factors may also contribute to gender-related differences in interpreting certain terms. Indonesian linguistic expressions related to anxiety or hopelessness sometimes carry gendered nuances, which could influence response patterns. These cultural-linguistic nuances may help explain why DIF emerges even though the items meet statistical fit criteria.

Although DIF was detected, the magnitude of DIF for most items was moderate rather than severe. This suggests that DIF does not necessarily invalidate the entire subscale but indicates that caution is needed in interpreting gender-based comparisons. Current psychometric recommendations emphasize evaluating the impact of DIF before removing items. In this study, DIF did not meaningfully distort overall subscale reliability or structure, therefore, item removal is not immediately warranted. Instead, items may be retained with caution or considered for revision in future adaptations.

Previous Indonesian studies, using CFA have not reported DIF (Widyana, 2020), likely because these methods do not detect item-level bias in the same way Rasch does. International Rasch-based studies, such as those conducted in Spain and China, have also documented gender-related DIF in DASS-21, particularly in the anxiety subscale (Wang *et al.*, 2025). Thus, the presence of DIF in the Indonesian version is consistent with global findings, suggesting that gender-related bias in emotional symptoms reporting may be a broader cross-cultural phenomenon.

Several limitations should be acknowledged. First, the use of convenience sampling limits the generalizability of findings, as the sample may not fully represent students from all regions or institutional types in Indonesia. Second, DIF analysis was limited to gender, other demographic variables such as socioeconomic status, academic discipline, or cultural background were not examined but may influence item functioning. Finally, because data were collected during a period of relative post-pandemic stabilization, distress levels may differ from those observed during acute pandemic periods.

Conclusion and Implication

Conclusion

This study concludes that 15 items of the Indonesian version of the DASS-21 were found to be valid and reliable in representing psychological distress. For future research, it is recommended that the six items violating the principle of fairness in measurement be carefully reconsidered, revised, or modified before being used again in research or mental health assessments.

Implication

The implications of this study highlight the need for further examination of the six items identified with Differential Item Functioning (DIF) to determine the source of misfit, such as potential issues in language adaptation, cultural context, or item interpretation by participants. Revision or modification of these six items is essential to ensure that the Indonesian version of the DASS-21 can be fully applied in the Indonesian context.

Declaration

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Author contributions

YN and F contributed to conceptualization, methodology, and writing of the manuscript; YN data analysis and editing.

Conflict of interest

We declare no conflicts of interest regarding the publication of this paper.

Ethical Clearance

This study was carried out in accordance with the recommendation of Code of Ethics of Scientific Psychology Consortium of Indonesia (Konsorsium Psikologi Ilmiah Nusantara) with No. 016/2022 Etik/KPIN.

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